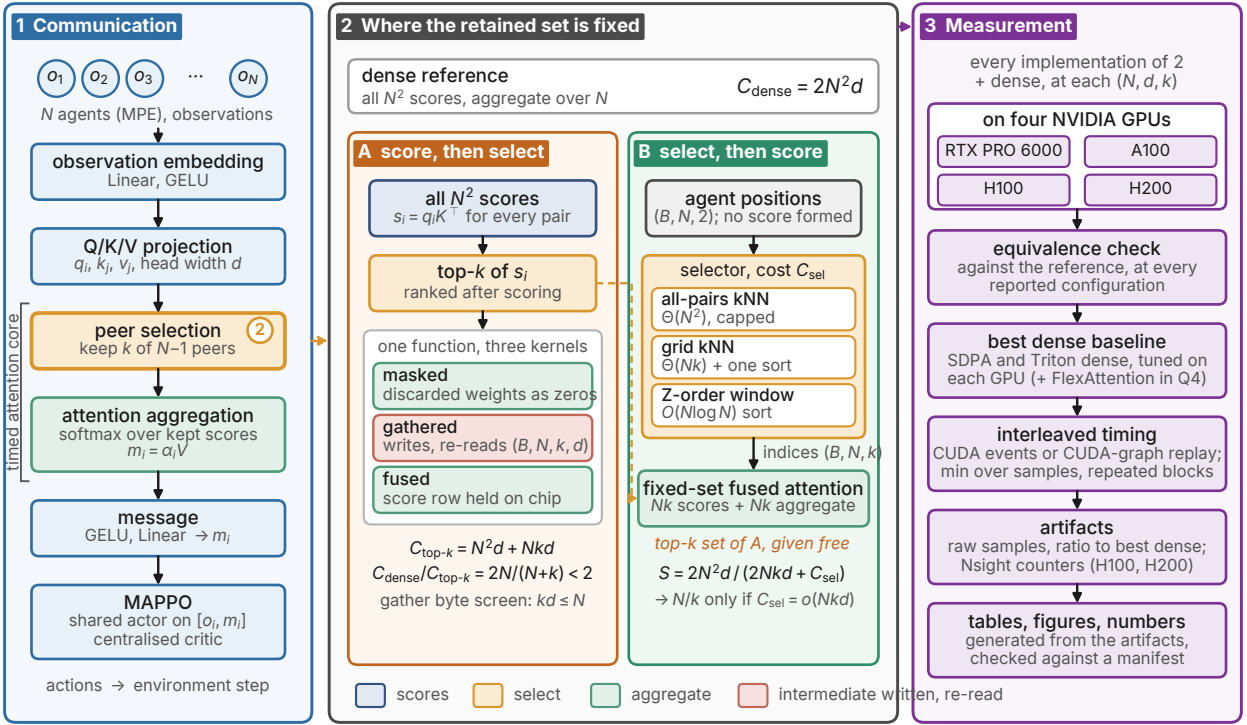

1 Communication
o_1 o_2 o_3 ... o_N
N agents (MPE), observations
observation embedding
Linear, GELU
Q/K/V projection
q_i, k_j, v_j, head width d
timed attention core
peer selection
keep k of $N-1$ peers
2
attention aggregation
softmax over kept scores
$m_i = \alpha_i V$
message
GELU, Linear → m_i
MAPPO
shared actor on $[o_i, m_i]$
centralised critic
actions → environment step
2 Where the retained set is fixed
dense reference
all N^2 scores, aggregate over N
$C_{\text{dense}} = 2N^2d$
A score, then select
all N^2 scores
$s_i = q_i K^\top$ for every pair
top-k of s_i
ranked after scoring
one function, three kernels
masked
discarded weights as zeros
gathered
writes, re-reads (B, N, k, d)
fused
score row held on chip
$C_{\text{top-}k} = N^2d + Nkd$
$C_{\text{dense}}/C_{\text{top-}k} = 2N/(N+k) < 2$
gather byte screen: $kd \le N$
B select, then score
agent positions
$(B, N, 2)$; no score formed
selector, cost C_{sel}
all-pairs kNN
$\Theta(N^2)$, capped
grid kNN
$\Theta(Nk)$ + one sort
Z-order window
$O(N \log N)$ sort
indices (B, N, k)
fixed-set fused attention
Nk scores + Nk aggregate
top-k set of A, given free
$S = 2N^2d \,/\, (2Nkd + C_{\text{sel}})$
→ N/k only if $C_{\text{sel}} = o(Nkd)$
scores
select
aggregate
intermediate written, re-read
3 Measurement
every implementation of 2
+ dense, at each (N, d, k)
on four NVIDIA GPUs
RTX PRO 6000
A100
H100
H200
equivalence check
against the reference, at every
reported configuration
best dense baseline
SDPA and Triton dense, tuned on
each GPU (+ FlexAttention in Q4)
interleaved timing
CUDA events or CUDA-graph replay;
min over samples, repeated blocks
artifacts
raw samples, ratio to best dense;
Nsight counters (H100, H200)
tables, figures, numbers
generated from the artifacts,
checked against a manifest